Stuck & Recovery Analysis

Endovascular navigation RL · run v1bp · explore-only

8,040	2,551,498	52.8%	10,999
episodes	explore steps	episode success	stall events

SCOPE. Every number in this report is measured from the v1bp worker-log complete ledger (EPISODE_START / STEP / EPISODE_OUTCOME), restricted to EXPLORE episodes. Evaluation episodes are excluded exactly: runner.eval() resets with explicit seeds, so eval EPISODE_START lines carry a `seed=` field and explore resets do not. That filter yields 8,040 explore episodes and exactly 980 eval (10 evals x 98) — exact. The obvious alternative, main.log time windows, leaks; and episode_summary.jsonl is unusable because it logs only ~36% of explore episodes. TRAINING-TIME AXIS. Cumulative explore steps, summed per-episode in chronological order—exact from the logs, unlike wall-clock (which conflates simulator speed) or the global step counter. THIS RUN. v1bp is byte-identical to v1b except for ONE change: it adds a reward pair (tip-average progress + catheter-slack potential). It is a single-variable experiment. SELF-CONTAINED. Every table and every claim in this report is derived from this run alone. Nothing here depends on comparison with another run.

Generated from monitoring/extract_stuck.py records · canonical detector config unless stated

Definitions — and why each one is drawn where it is

Every downstream number depends on these choices, so they are stated before any result

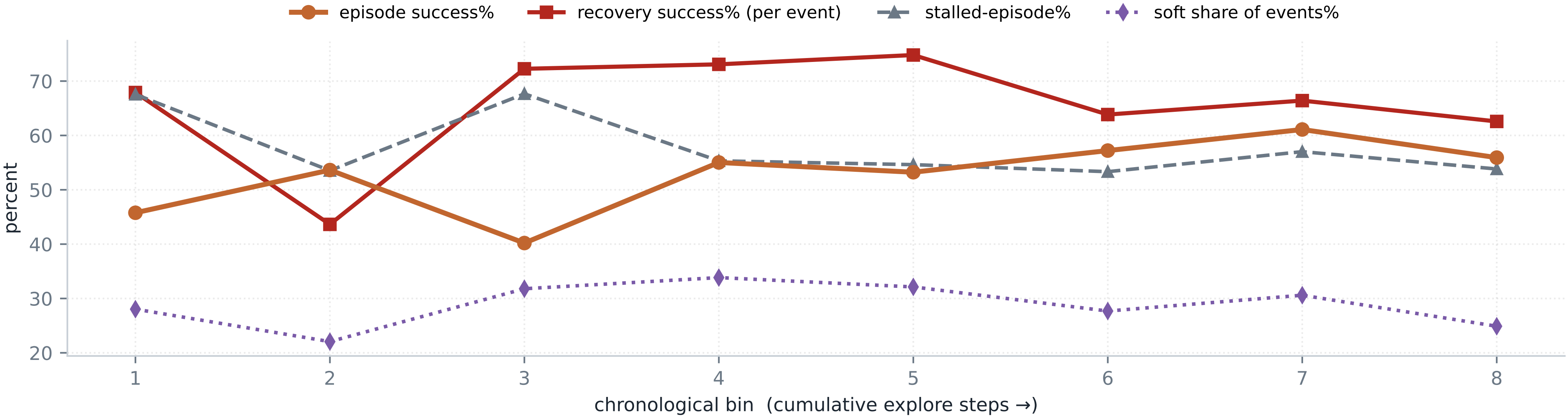
term	unit	definition
stall event	per event	wire pushing (|gw_trans| > 2 mm/s) while path progress stays flat (< 0.3 mm gain) for 12 consecutive steps
grind	per event	escaped the stall after < 1 mm of withdrawal — forced through
soft	per event	escaped after 1–8 mm withdrawal — ease slack, re-advance (the manoeuvre we set out to induce)
hard	per event	escaped after > 8 mm withdrawal — costly pullback
unrecovered	per event	never passed the jam; terminal for that episode
recovery success%	per EVENT	(soft + hard + grind) / all stall events
episode success%	per EPISODE	reached the target

A stall is 'pushing but not progressing', not merely 'not progressing'. A wire that is deliberately retracting or rotating is manoeuvring, not stuck. Requiring the push command isolates the case where the policy IS trying to advance and the vessel will not allow it. The stall counter decays by 2 per non-stalled step rather than resetting, so one noisy command sign-flip inside a genuinely stuck push does not erase the evidence. Recovery kind is classified by ACTUAL withdrawn length — peak insertion minus deepest insertion during the stall — not by the commanded retraction. A retract command that moves nothing (wire wedged) must not score as a soft recovery. THE THREE SUCCESSES sit at different units of analysis, which is the single most important thing on this page: (1) EPISODE success is per episode — did it reach target. (2) RECOVERY success is per EVENT — did the wire escape a given stall. (3) RECOVERY-TO-EPISODE success is per episode — did escaping actually convert into reaching the target. Conflating (1) and (2) makes 'success tracks recovery' look true when the opposite can hold.

Stuck & recovery vs training time

Each bin ~ 1800 episodes of training data

bin	explore steps (k)	success %	stalled-ep %	events	recovery %	soft %	hard %	grind %	unrecovered %
1	0–374	46%	68%	1623	68%	28%	14%	26%	32%
2	375–671	54%	54%	825	**44%**	**22%**	**11%**	**11%**	**56%**
3	671–1060	40%	68%	1947	72%	32%	22%	18%	28%
4	1061–1370	55%	55%	1501	73%	34%	18%	21%	27%
5	1370–1686	53%	55%	1603	75%	32%	21%	22%	25%
6	1686–1979	57%	53%	1181	64%	28%	17%	19%	36%
7	1979–2257	**61%**	57%	1170	66%	31%	15%	21%	34%
8	2257–2551	56%	54%	1149	63%	25%	15%	22%	37%



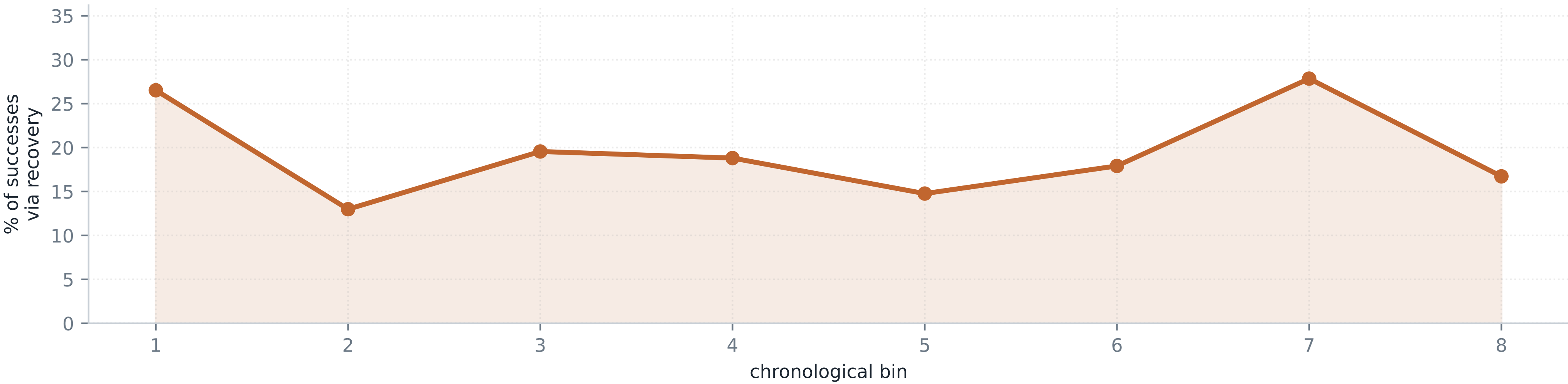
WHAT IT SHOWS How often the wire got stuck, how often it got out, and by which manoeuvre, as training progressed, for the run carrying the reward pair.

INFERENCE Recovery does NOT decay in this run. It holds 63–75% from bin 3 onward, and the stalled-episode share stays roughly flat (53–68%) rather than falling away. The reward pair kept the recovery skill alive, exactly as designed. But episode success moves only 46→56% across the whole run: the policy keeps entering stalls and keeps fighting out of them, and the improved escape ability does not translate into a materially better outcome. The later pages show where that escape effort is lost.

Success vs Recovery

Each episodeassigned to exactly one channel: clean / stuck-and-recovered / stuck-and-failed

bin	success %	clean eps (success %)	recovered eps (success %)	unrec eps (success %)	% of successes via recovery
1	46%	326 (100%)	158 (77%)	521 (2%)	27%
2	54%	467 (100%)	73 (96%)	465 (0%)	13%
3	40%	325 (100%)	140 (56%)	540 (0%)	20%
4	55%	449 (100%)	152 (68%)	404 (0%)	19%
5	53%	456 (100%)	145 (54%)	404 (0%)	15%
6	57%	469 (100%)	109 (94%)	427 (1%)	18%
7	61%	432 (100%)	180 (95%)	393 (3%)	28%
8	56%	464 (100%)	111 (85%)	430 (1%)	17%



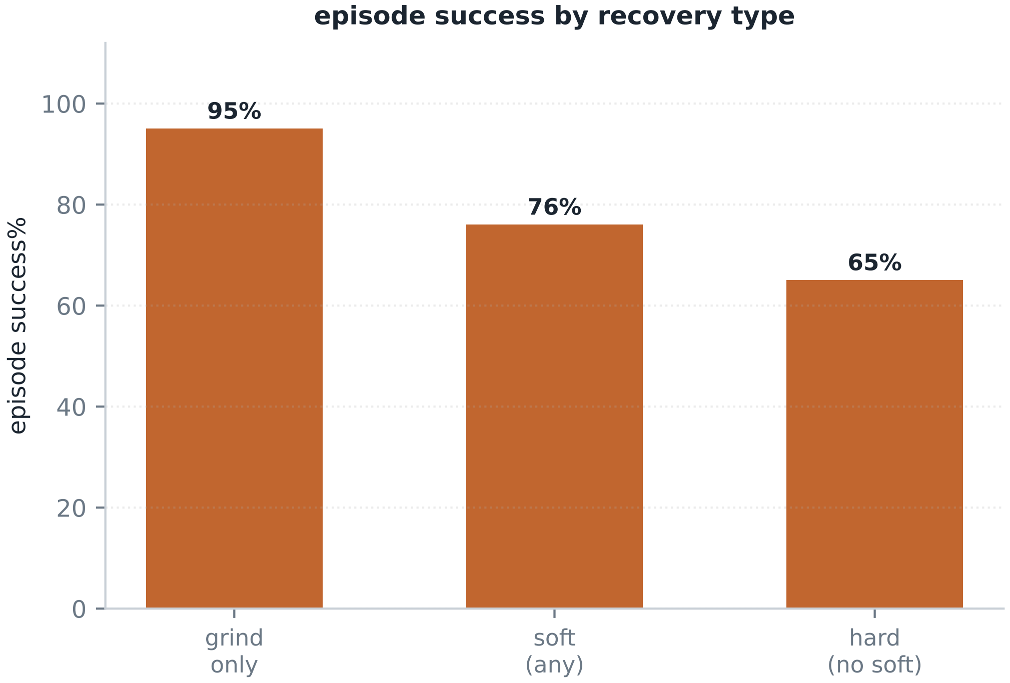
WHAT IT SHOWS   Episode success decomposed by channel, and the share of all successes that flowed through recovery (chart).

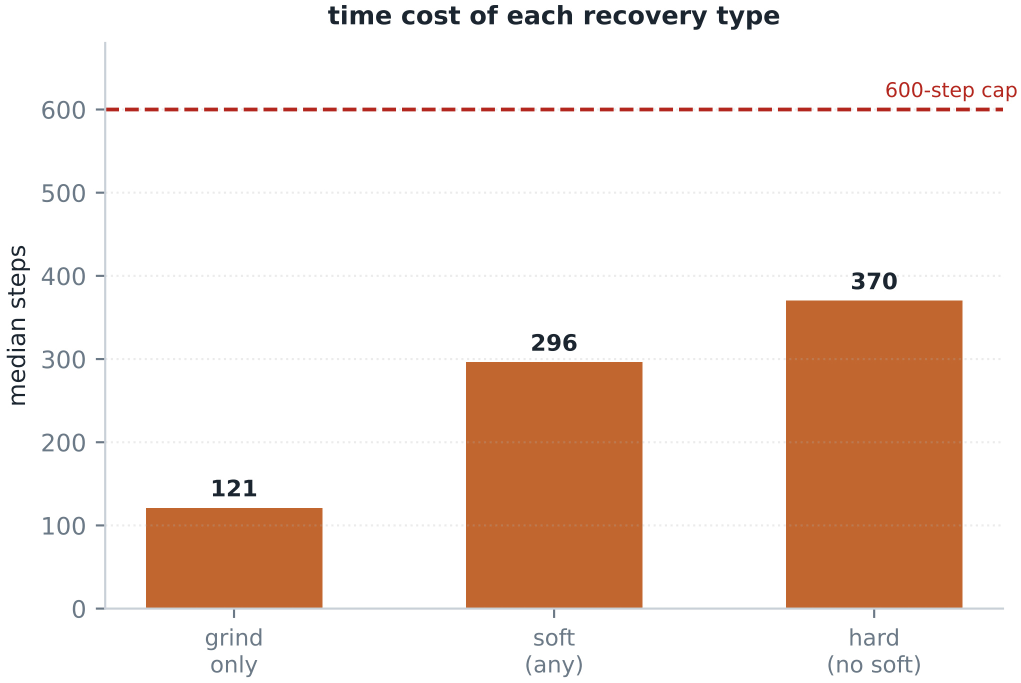
INFERENCE   No downward trend: recovery's contribution stays in the 13–28% band across the whole run, and the best-performing bin is among those leaning MOST heavily on recovery. This run never migrates away from the recovery channel — it keeps reaching the target by fighting through stalls rather than by avoiding them, right to the end of training. Note: As seen by the success rates in columns 4 and 5, a small fraction of episodes that recover do not reach the target as they have exhausted all their steps and similarly some that do get stuck are already very close to the target and hence gets counted as successes.

Recovery type vs episode success

Episodes grouped by the recovery they achieved

recovery profile	success %	median steps	mean retract mm
soft (any)	76%	296	6.2
hard (no soft)	65%	370	15.2
grind only	**95%**	**121**	0.1
unrecovered (any)	1%	600	NA
never stalled	100%	43	0.0





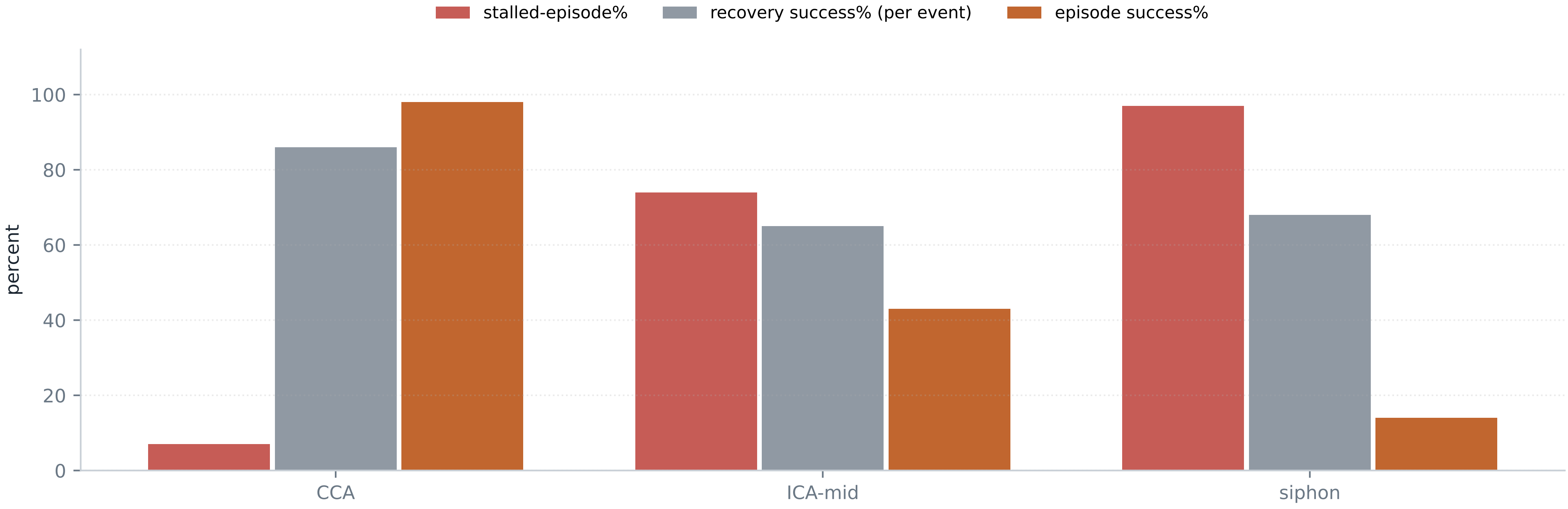
WHAT IT SHOWS   Whether the type of escape predicts the episode outcome, and what each type costs in steps.

INFERENCE   GRIND BEATS SOFT: grind-only episodes succeed 95% in 121 median steps, against soft at 76% in 296 steps and hard at 65% in 370. The likely reason is a confound: recovery type is mostly a proxy for stall SEVERITY, not policy skill. A jam you can grind through was mild; one forcing an 8 mm withdrawal was severe and usually deeper, where the episode was already in trouble.

Stuck & recovery by depth

Path-length bands: CCA (<146 mm) · ICA-mid (146–210 mm) · siphon (>210 mm)

band	eps	ep succ%	stalled-ep%	events	recovery succ%	soft%	hard%	grind%	unrec%
CCA	2804	98%	7%	457	86%	36%	18%	32%	14%
ICA-mid	2690	43%	74%	4217	65%	30%	17%	19%	35%
siphon	2546	**14%**	**97%**	6325	68%	29%	18%	21%	32%

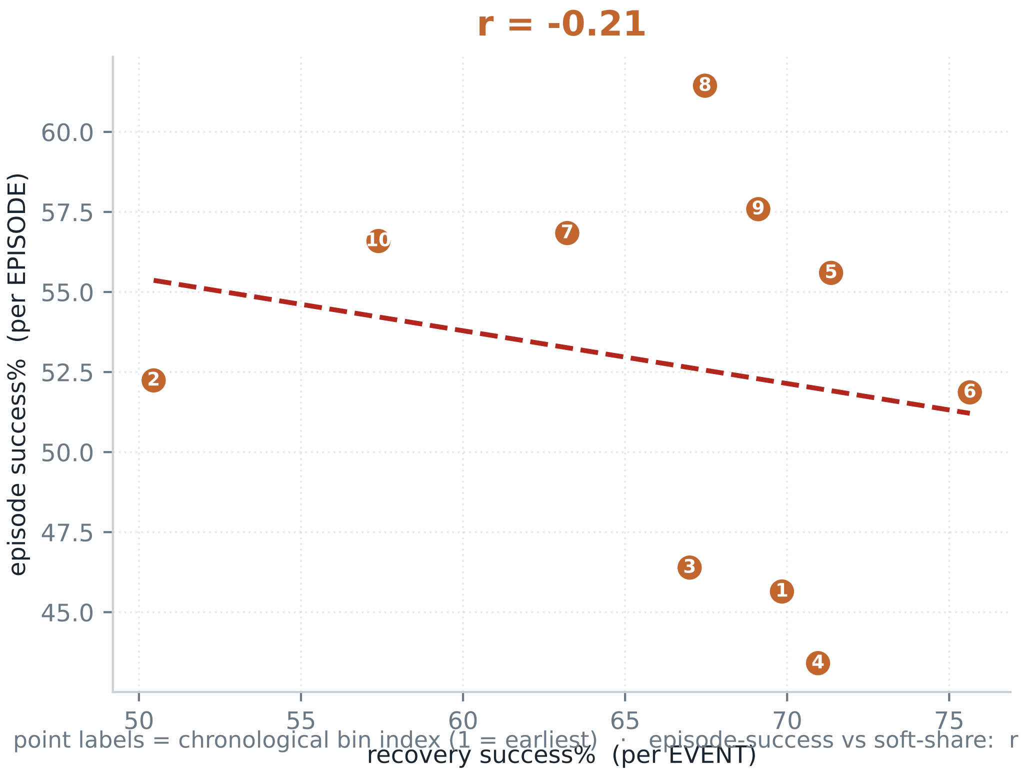


WHAT IT SHOWS   Where the stalls actually are, and whether escaping them rescues the episode there.

INFERENCE   Depth dominates every other factor. Stalling runs 7% at the CCA but 97% at the siphon — getting stuck is essentially guaranteed once the wire enters the cavernous segment. And escaping does not save you there: 68% of siphon stalls are escaped yet only 14% of siphon episodes succeed. You escape one jam and immediately meet the next. This reframes the siphon as a STALL-DENSITY problem rather than an escape-skill problem — the same wall the real-patient evaluation hit at 0 of 30, seen from the inside.

Is episode success actually tracking recovery?

10 chronological bins · each point is one bin · units deliberately mixed to test the intuition



point labels = chronological bin index (1 = earliest) · episode-success vs soft-share: r = -0.07

WHAT IT SHOWS A direct test of the intuition that success rises and falls with recovery skill. Each point is one chronological bin; the dashed line is the least-squares fit.

INFERENCE Essentially NO relationship (r = -0.21) — and that is itself the finding. Episode success and recovery success move independently here: recovery ability neither drives the outcome nor has to decay for the outcome to improve. Statistically this is what a preserved but non-decisive recovery skill looks like — the two quantities are simply not the same thing, which is why they are separated by unit of analysis in T2.

Threshold sensitivity — are these conclusions detector artifacts?

Every episode scored under three detector configurations simultaneously

v1bp under three detectors

config	events	stalled-ep%	recovery succ%	soft%	hard%	grind%	unrec%
sens	14,271	61%	75%	36%	16%	23%	25%
canon	10,999	58%	67%	30%	17%	21%	33%
strict	8,701	55%	60%	26%	18%	15%	40%

detector parameters

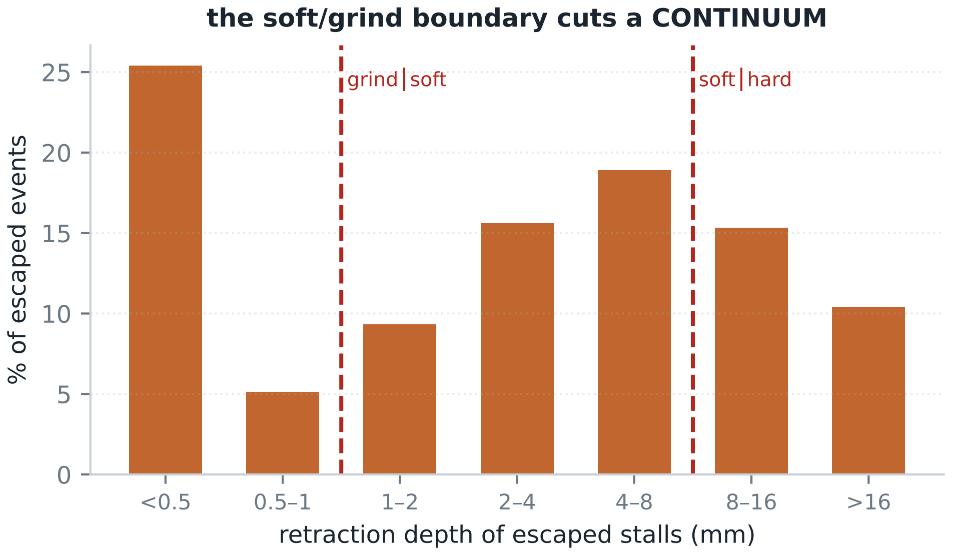
config	stall_eps	push_min	stuck_steps	retract_min	soft_max	pass_eps
sens	0.3 mm	1.0 mm/s	8	0.5 mm	8 mm	1.0 mm
canon	0.3 mm	2.0 mm/s	12	1.0 mm	8 mm	1.0 mm
strict	0.5 mm	4.0 mm/s	16	1.0 mm	8 mm	1.0 mm

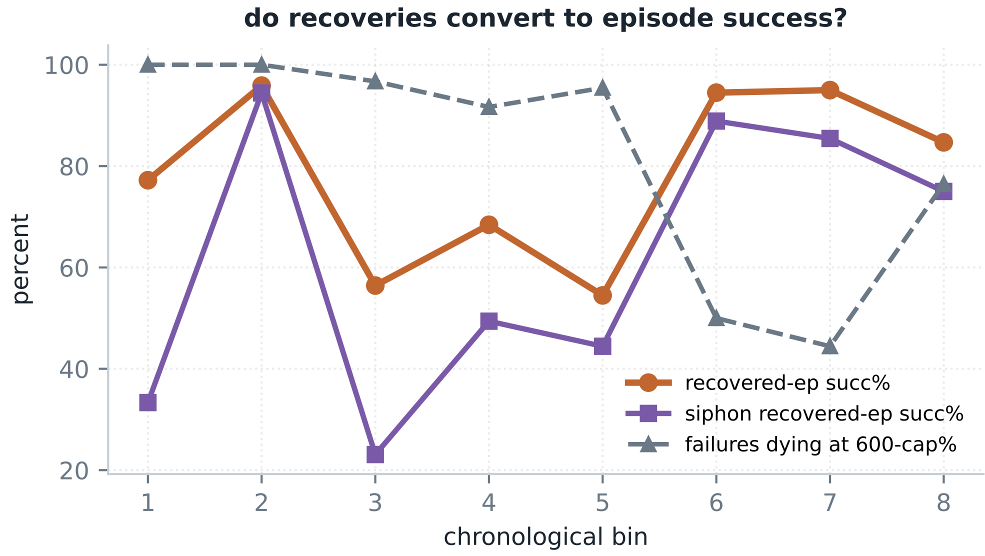
WHAT IT SHOWS The same analysis re-run under a sensitive, a canonical and a strict definition of 'stuck', to separate signal from detector choice.

INFERENCE Absolute rates move a great deal — recovery success reads anywhere from 60% to 75% for this run alone depending on strictness — so NO absolute number in this report should be quoted on its own. What survives all three settings is the ORDERING and the DIRECTION of every trend, which is why each finding here is stated as a delta or a trend rather than a level.

Diagnostics — recovery quality, cost and where it fails

Retraction-depth distribution, per-bin conversion, and the forensics of recovered-but-failed episodes





conversion of recovered episodes, by depth

band	recovered eps	episode succ%
CCA	144	94%
ICA-mid	503	85%
siphon	421	61%

what the failures were doing

recovered-but-FAILED episodes	value
count	246 of 1068 recovered
died at the 600-step cap	91%
median final distance to target	61 mm

WHAT IT SHOWS   Left: the retraction-depth distribution of escaped stalls, with the classification cuts drawn on. Right: how well this run's recoveries convert to episode success over training. Below: conversion by depth band and what the failed recoveries were doing when they ended.

INFERENCE   The retraction distribution is a CONTINUUM massed near zero (25% of escapes are under 0.5 mm), not a bimodal soft/grind split — so the 1 mm and 8 mm cuts slice a smooth curve and absolute soft% is detector-defined rather than a property of the policy. Recovered episodes convert at only 77%, and the failures TIME OUT rather than jam — 91% die at the 600-step cap, a median 61 mm short of target. They escaped every stall and simply ran out of clock. Note the conversion is strongly depth-dependent and largely TRANSIENT: the late bins recover to the 85–95% range, so this run did eventually learn to make recoveries pay.

Findings

Every claim is stated as a delta or trend, and survives all three detector configurations

F1 The reward pair fixed recovery — mechanically and unambiguously.

Recovery success holds 63–75% from bin 3 onward instead of decaying, unrecovered sits at 32.6% of events, soft share is 29.5%, and mean catheter-lead — the shove pathology the pair targeted — is 0.04. The pair did exactly what it was designed to do.

F2 …and the escape ability does not convert into success.

Episode success ends at 52.8% overall despite the strong escape numbers. Recovered episodes convert at only 77%, and their failures TIME OUT rather than jam — 91% die at the 600-step cap a median 61 mm short of target, having escaped every stall along the way. Escaping is necessary and nowhere near sufficient.

F3 Recovery is preserved but non-decisive, and the conversion deficit is TRANSIENT.

Episode success and recovery success are statistically uncoupled ($r = -0.21$): recovery never withers, but neither does it drive the outcome. Conversion is also not a fixed property — it recovers to 95% in the late bins, so the policy did eventually learn to make its recoveries pay. What it never learned is to stop entering stalls: the stalled-episode share ends at 54%, essentially where it started.

F4 GRIND still beats SOFT on episode success.

Grind-only 95% in 121 median steps vs soft 76% in 296 and hard 65% in 370. The ordering survives the reward pair, which strengthens the severity-confound reading: it is a property of the stalls, not of the gait.

F5 Depth dominates everything.

97% of siphon episodes stall versus 7% at CCA, and at the siphon 68% of stalls are escaped yet only 14% of episodes succeed. Better escape skill does not break the siphon — the limiting factor is stall DENSITY.

Implications

What these measurements license, and what they rule out

1 **The reward pair is validated as a MECHANISM and rejected as a WIN.**

It moved the intended internal behaviour decisively and produced no outcome gain. This is the cleanest evidence in the program against further reward shaping as the lever, and a publishable negative result in its own right.

2 **Better recovery is not the bottleneck.**

This run escapes two thirds of its stalls, leaves few terminal jams, and eliminates the catheter shove— and still finishes at the same order of success. Whatever caps performance sits above the choice of gait.

3 **Recovery must be CHEAP, not merely successful.**

The failures here escaped every stall and still lost — on the clock, tens of millimetres from target. Under a fixed step budget, escape SPEED matters as much as escape ability, and a recovery that consumes half the episode is close to worthless.

4 **Pre-register OUTCOME criteria, not behaviour criteria.**

Every leading indicator this intervention was designed to move did move, decisively and in the intended direction, while the outcome did not follow. Behavioural evidence that a change 'worked' is not evidence that it helped.

Reproduce: `monitoring/extract_stuck.py <v1bp_run_dir> stuck_v1bp.jsonl` then `monitoring/report_single.py stuck_v1bp.jsonl v1bp <out.pdf>`

All figures in this report are derived from this run alone.